Nicholas Chin Bo You

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Education

University of Utah *August 2024 – Present*
Salt Lake City, UT

BS in Physics with Computational Physics and Astronomy Emphasis
BS in Applied Mathematics

- CGPA: 3.872
- Dean's List Fall 2024, Spring 2025, Fall 2025, Spring 2026, Summer 2026

Foon Yew High School *Jan 2018 - Nov 2023*
Johor Bahru, Malaysia

Publications

Measurement of the photosphere oblateness of γ Cassiopeiae via Stellar Intensity Interferometry with the VERITAS Observatory
*A. Archer, J.P. Aufdenberg, . . . , **N. B. Y. Chin**, . . . , The VERITAS Collaboration*
The Astrophysical Journal, 995, 191 (2025)

Shared Risk-Capital Constraints in Dynamic Option Dealer Hedging
Submitted to SIAM Journal on Financial Mathematics

Research Experiences

Analysis of Extended Gamma Ray Sources with VERITAS Using Gammapy
University of Utah Gamma Ray Astronomy Research Group | Advisor: Prof. David Kieda | Summer 2026 - Present

- Investigating two-dimensional spatial and spectral analysis of Cassiopeia A using Gammapy, including morphology and spectral model fitting.
- Implemented a gamma ray extended source analysis pipeline for VERITAS observations.
- *2026 4CS Poster Presentation*

Verification of VERITAS Results Using the Gammapy Analysis Package (SURF Program)
University of Utah Gamma Ray Astronomy Research Group | Advisor: Prof. David Kieda | Summer 2026

- Implemented a gamma ray data analysis pipeline using Gammapy for VERITAS observations and validated its results through independent comparisons with analyses performed by researchers working on the VERITAS–Gammapy integration.
- Conducted point-source analyses of the Crab Nebula and Cassiopeia A, including background estimation, significance mapping, and spectral modeling to characterize their very-high-energy gamma-ray emission.
- *Summer Symposia Presentation*

Efficient Quantum State Preparation Algorithms for Astronomical Image Encoding
Independent Research | Advisor: Prof. Massood Tabib-Azar | Spring 2026 - Present

- Investigating efficient quantum state preparation algorithms for encoding astronomical images on quantum computers, by exploiting image sparsity and nonzero structure to reduce quantum circuit complexity.
- Implemented several quantum state-preparation algorithms for Quantum Probability Image Encoding (QPIE), including methods designed to exploit sparsity and structured data.
- Benchmarked the algorithms using circuit complexity metrics and tested the fidelity of the prepared quantum states against their target states.

Stellar Intensity Interferometry: Search of Modulation Signatures in Variable Star Measurements and Background Analysis
VERITAS Stellar Intensity-Interferometry Research Group | Advisor: Prof. Tugdual LeBohec & Prof. David Kieda | Fall 2024 - Spring 2026

- Refined and maintained Stellar Intensity Interferometry (SII) data analysis workflows using CERN ROOT and C++, improving the processing and interpretation of observational data for the University of Utah SII research group.

- Analyzed observational data from the β Cephei variable β Canis Majoris to investigate temporal modulation signatures associated with stellar photospheric pulsations.
- Prototyped a data format converter between FPGA-correlated data and ROOT-compatible data structures for integration with the existing analysis framework.
- Performed supporting computations to test and evaluate a moonlight-induced background model for SII observations.
- *2025 4CS Poster Presentation*

Presentations

APS Four Corners Section 2026 Annual Meeting	Fall 2026
University of Utah Undergraduate Research Symposium - Summer Symposia	Summer 2026
Quantum Computing Focused Reading Group	Spring 2026
◦ Affliated with NSF Research Training Group	
APS Four Corners Section 2025 Annual Meeting	Fall 2025
University of Utah Department of Physics and Astronomy Annual Symposium	Spring 2025

Skills

- Languages: English, Mandarin Chinese, Cantonese Chinese, Hakka Chinese, Bahasa Melayu
- Programming Languages: Python, C/C++, Rust, Java, Bash, LaTeX
- Scientific Computing : CERN ROOT, Gammapy, Qiskit, Jupyter Notebook, NumPy, Matplotlib, MATLAB
- Operating Systems: Linux / Unix

Awards and Certificates

Simons Laufer Mathematical Sciences Institute – SACNAS Travel Award	Summer 2026
Summer Undergraduate Research Fellowship (SURF)	Summer 2026
Thomas J. Parmley Scholarship Endowment Fund	Fall 2026 - Spring 2027
MIT iQuHACK 2026 - BlueQubit Problem	World Rank 35, 2nd Highest Score
Undergraduate Research Opportunity Program Award (UROP)	Spring 2026
Talent Scholarship	Fall 2025 - Spring 2026
Chen Jingrun's Cup Secondary School Mathematics Competition 2021	National Rank 81st
Chen Jingrun's Cup Secondary School Mathematics Competition 2023	High Distinction
OpenAI Codex Hackathon	1st runner up
IELTS 2024	6.5
MATLAB Onramp	Course Completion Certiicate

Other Experiences

Research Assistant — *U of U*
University of Utah Astroparticle Physics | Gamma Ray Astronomy Research Group — *Nov 2024 – Present*

- Working with Prof. Tugdual LeBohec and Prof. David Kieda on Stellar Intensity Interferometry
- Working with Prof. David Kieda on Gamma Ray Astronomy

Lead Organizer - Qiskit Fall Fest 2026 — *Salt Lake City, UT*
IBM Quantum — *Aug 2026 – Present*

- Selected by IBM Quantum from 900+ applicants to lead one of the 200 Qiskit Fall Fest events at the University of Utah.
- Coordinating a team of student organizers and faculty supporters to develop quantum computing workshops, invited talks, community activities, and a student hackathon.

Chair - Undergraduate Student Advisory Committee — *U of U*
U of U Physics Department — *Oct 2025 – Present*

- Expanded committee membership by 300% through active recruitment and outreach initiatives.
- Organized a department-wide social event in collaboration with the Society of Physics Students (SPS), attracting 30+ participants across undergraduate, grad-

uate, and non-physics programs.

- Led a student town hall attended by 40+ students to collect and communicate departmental feedback to faculty leadership.

Officer - Blockchain Club *U of U*
Treasurer - Fintech Club *Jan 2025 – Jun 2025*
U of U Finance Department *Jul 2025 – Present*

- Developed educational content introducing blockchain, frontier science, and emerging computational technologies.
- Coordinated the inaugural FinTech Career Conference and developed budget models to support event planning and financial management.
- Authored educational articles for the club's official website.

CFO - UCubesat Club *U of U*
U of U ECE Department *Jan 2025 – May 2025*

- Telemetry Team; Ground Station Subsystem Team Lead.
- Working on building a ground station to contribute to the SatNOGS Network.

Engineering Possibilities Summit Scholar *Jan 2025 - Dec 2025*
Goldman Sachs

President - Badminton Club *Sept 2026 - Present*
U of U Eccles Student Life Center

Apprentice Management *Johor Bahru, Malaysia*
Rhapsody Youth Ensemble *2 projects (Approximately 6 months)*

- Established the foundational framework for ensemble's operations and concert planning.

Co-Chairman and President of Education *Johor Bahru, Malaysia*
Foon Yew Symphony Orchestra *June 2022 - May 2023*

- Utilized recording and mixing skills to support the orchestra during the COVID period, facilitating virtual performances
- Organized and conducted more than 10 hours of online courses on different subjects weekly to maintain member engagement and skill development.
- Designed programs to help students foster interest and performing opportunities. Approximately 80% of members increased 40% more performing opportunities annually

Assistant Luthier and Sales Assistant *Johor Bahru, Malaysia*
Melodia String *Jan 2024 - Aug 2024*

- Performed walk-in instrument maintenance and basic repairs, including string replacement, setup adjustments, and minor restorations
- Studied and completed projects up to an intermediate level luthiery techniques
- Maintained workshop equipments and tools

Recording Studio Assistant *Johor Bahru, Malaysia*
Goldstone Music *Jan 2024 - Aug 2024*

- Assisted with recording session setup, including microphone placement, signal routing, and basic audio equipment configuration
- Provided technical and logistical support during live stage performances and events
- Studied music arrangement and audio recording techniques through hands-on observation and guided practice

Projects

Option Pricing using Quantum Computers *Option-Pricing-on-Quantum-Computer*

- A from-scratch implementation of Stamatopoulos et al. (2020). "Option Pricing using Quantum Computers"

A Guide to Quantum Computing *A-Guide-to-Quantum-Computing*

- A collection of notes and implementations aimed at building intuition for quantum computing concepts without requiring prior knowledge of quantum mechanics.

A Complete DNS Server *DNS-Server-in-RUST*

- Developed a fully functioning DNS Server that handles DNS queries, resolves domain names, and returns IP addresses following standard DNS protocol specifications with Rust.

Relevant Coursework

Thermodynamics and Statistical Mechanics	In Progress
Quantum Mechanics and Electrodynamics	In Progress
Intermediate Mechanics	In Progress
Partial Differential Equations	In Progress
Physics II Lab	In Progress
Foundations of Analysis I	A-
Intro to Probabilities	A
Intro to Quantum Computers - Graduate Course	A
Enhanced Intro to Differential Equations - Honors Course	B
Modern Physics	A
Survey of Numerical Analysis	A
Enhanced Linear Algebra - Honors Course	B+
Foundations of Astronomy	A
Accelerated Object-Oriented Programming	A-
Physics I, II	A
Physics I Lab	A
Calculus I, II, III	A